Gas Turbine Single Annular Combustor Sector: Combustion Dynamics

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This paper investigates the combustion dynamics of a single cup gas turbine combustor sector, a swirl-stabilized burner with dilution and cooling air holes. The burner was fueled by natural gas and was operated at one atmospheric pressure with various air inlet temperatures, air pressure drops, and air–fuel ratios. Acoustic emissions as functions of operating conditions were measured. Chemiluminescence imaging with a high-speed CCD camera and simultaneous dynamic pressure measurements has been used to characterize the combustor's behavior. Imaging of the combustion energy release rate has provided an insight on the flame structure and its interaction with the dilution air jets. The combustion dynamics were correlated well with the heat release rate in the primary zone or in the dilution air jet region. Two low-frequency combustion dynamics modes (fuel-rich combustor domes with frequency around 420Hz, and fuel-lean combustor domes with frequency around 280 Hz) were identified by a dynamic microphone, dynamic pressure transducers, and a high-speed CCD camera. A two-dimensional distribution of the heat release rate dynamics (amplitude and phase angle) reveals different combustion dynamics driving mechanisms for fuel-lean and fuel rich modes. The acoustic emissions for the fuel-lean combustion mode correlated well with the energy release rate dynamics of the swirling jet inside the combustor dome. The acoustic emissions for the fuel-rich mode were associated with the unsteady chemical energy release rate from interactions between the dilution air jets and the fuel-rich mixture convected from the combustor dome.

Nomenclature

f	=	frequency of oscillation
LPP	=	lean premixed combustion
RQL	=	rich-burn, quick-mix and lean-burn
T	=	temperature
ΔP	=	pressure drop at swirler
ΔT	=	inlet temperature step
Φ	=	equivalence ratio
p'	=	acoustic pressure

I. Introduction

The concept of premixed combustion (LPP and RQL) has been widely applied in the industry, particularly in gas turbine industry, as an effective means to achieve high combustion efficiency and meet the stringent environmental standards of NO*x* emissions. This type of combustion exhibits a wide range of combustion dynamics, induced by coupling between unsteady heat release and acoustic oscillations. This is often manifested as high magnitude acoustic noise, known as thermo-acoustic instabilities, which were first investigated by Rayleigh [1]. The

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occurrence of combustion dynamics in gas turbine combustors presents a great challenge for engineers. Not only are combustion dynamics detrimental to the operation of the engine and combustor, but the difficulty in predicting and remedying dynamics problems can lead to significant financial loss, and delays in engine development. Therefore, thermo-acoustic instabilities must be considered in the early stages of the combustion system design phase.

A variety of complex physical processes may be involved in the combustion system stability. Extensive experimental and theoretical works have been conducted to identify the fundamental mechanisms and to devise analytical models. Various mechanisms have been founded to initiate thermo-acoustic instabilities. For instance, air–fuel ratio oscillations, induced by oscillating pressure, found at the combustor inlet (Crocco et al [2,3]), may couple with the heat release oscillations and drive the instability. Interactions of the acoustic velocity oscillations with the flame cause periodic flame front variation, which induces a periodic heat addition process (Candel [4]). Variations of turbulence intensity (Schuermans et al [5]), atomization, vaporization, and mixing [2, 3] could also result in a periodic supply of fuel to the flame and an oscillatory equivalence ratio that produce heat release oscillations. Vortex shedding activity (Paschereit et al [6-9]) may distort the flame front and cause its surface area to oscillate, thus producing an oscillatory heat-release process that also drives the acoustic field. In gas turbine combustion, the feed-coupling mechanism leading to air–fuel ratio oscillations (Lieuwen et al[10]) and hydrodynamic instabilities have been identified as the most likely drivers for combustion noise. Although these examples illustrate the various mechanisms capable of driving combustion noise, the amount of energy fed back into the initial perturbation is modulated by the resonant acoustics of the combustion systems. Promoting the growth of the natural frequency modes of the combustion system, is one of the few characteristics common to all thermoacoustic instabilities (Lieuwen et al [11]).

While the global mechanisms of thermo-acoustic instability have been extensively studied, identifying the initial mechanism remains a difficult task. In all reality, several physically distinct mechanisms can drive combustion noise for the same combustor, depending on combustor's operating conditions. The composition and temperature of the mixture entering the combustion chamber, the swirl intensity, and the pressure, among others, are functions of operating conditions, and can determine the flame temperature, the flame front location, and the flame shape, which all have a direct impact on the flame's susceptibility to external perturbations. Although this topic has been widely studied by scientists and engineers, the added geometric complexities and injector configurations of a practical combustor make their dynamic behavior currently unpredictable. Current industry design techniques are largely empirical and not clearly defined with respect to combustion instabilities.

To improve our understanding of stability properties in such complex systems encountered in many industrial applications, the flame structure, of an atmospheric swirl-stabilized burner that included dilution and cooling air holes and was fed with natural gas fuel, was systematically investigated for various inlet temperatures, pressure drops, and air–fuel ratios. Chemiluminescence imaging with a high-speed CCD camera and simultaneous dynamic pressure measurements has been used to characterize the combustor's dynamic behavior. By changing the test conditions like the equivalence ratio, inlet temperature and pressure drop at the swirler, a complete map of the combustion pressure dynamics and acoustic emissions were established. By analyzing the high-speed camera images, the mechanisms of combustion dynamics have been identified.

II. Experiment setup and measurement methodology

The experimental measurements were conducted in an atmospheric combustion test rig that simulates a Single Annular Combustor (SAC) Sector. Figure 1 schematically shows the test facility employed in the present work. The test rig consists of an air inline heater, flow conditioning chamber, and the combustor. The test rig was oriented horizontally with the combustion air preheated by a 72 kW inline electrical heater to the desired temperature. Compressed air was delivered to the combustor by a compressor with the maximum mass flow rate of 1 kg/s during the test. A Micro-motion CMF 300 flow meter upstream of the heater measures the air flow rate. The flow conditioning chamber is constructed from 6 inches of schedule 40 stainless steel piping, with an inner diameter of 154.1 mm (6.065 inch) and a total length of 610mm (24 inch). Two PCB 112A05 type dynamic pressure transducers were mounted on the wall to measure the dynamics pressures in the flow conditioning chamber. A pipe expander, 152.4 mm (6 inch) inlet diameter and 203.2mm (8 inch) outlet diameter, was used to connect the air settling chamber to the combustor flange. A type K thermocouple and a static pressure transducer were used to measure the burner inlet air temperature and pressure.

Both liquid and gaseous fuels have been used in the test but only results from natural gas will be discussed in this paper. The natural gas fuel feed system includes a high pressure gas cylinder bank with maximum gas pressure of 1500 Psig. A pressure regulator was used to maintain the constant fuel supply pressure and a micro-motion CMF 25 flow meter to was used acquire the gas flow rate, which was controlled by a Parker precision valve. A type-K thermocouple and a Meriam pressure gauge were used to measure the fuel inlet pressure and temperature. All test conditions were acquired by a computer.

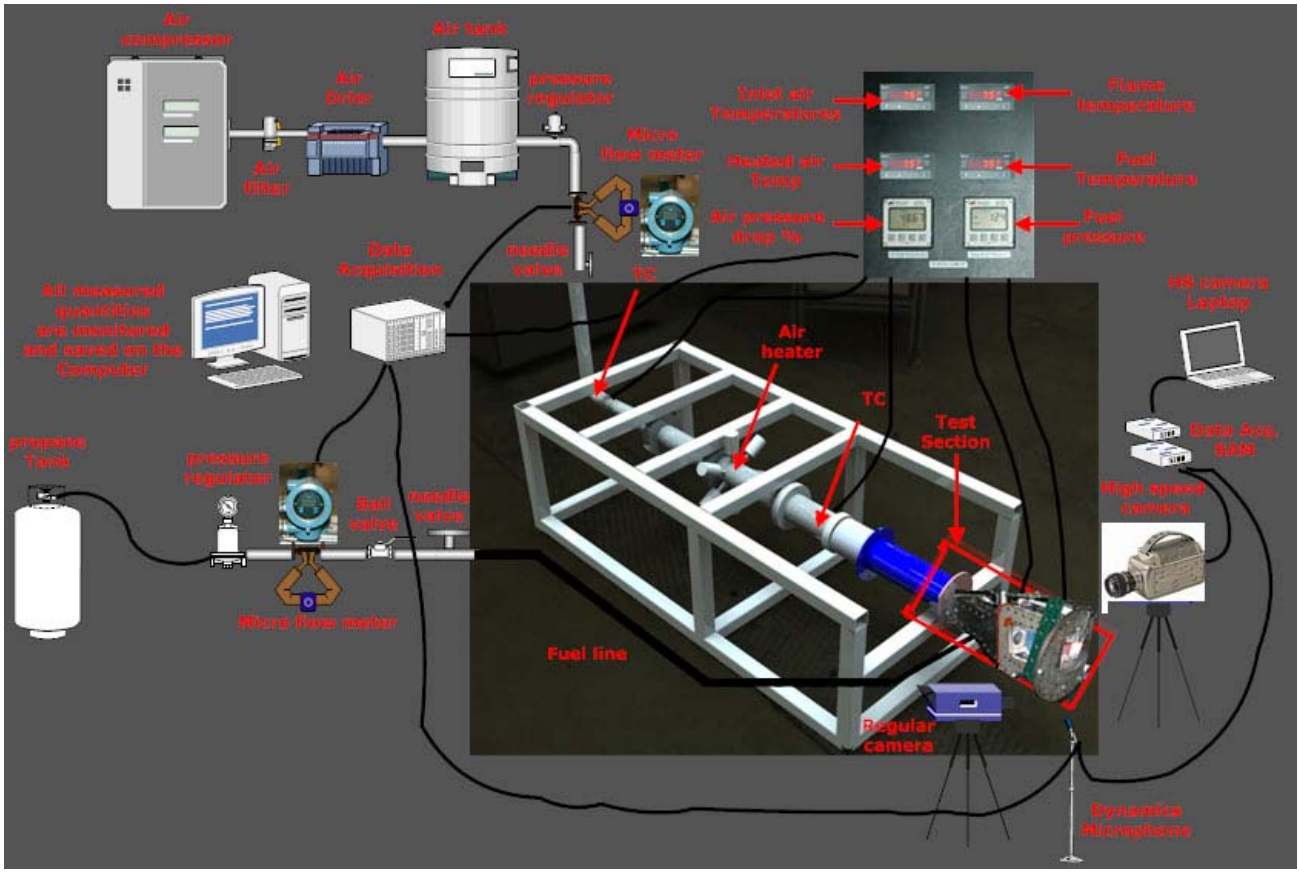
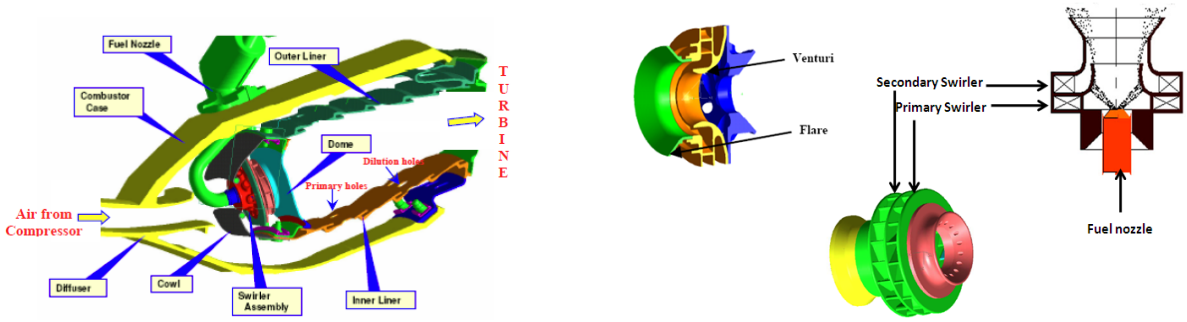


Figure 1: Experimental arrangement



(a) (b)

Figure 2: (a) Single Combustor Sector and (b) swirler assembly

Figure 2 is a generic sketch of the Single Annular Combustor Sector used in this investigation. The swirler cup features two radial inlet swirlers that have counter-rotating directions, and a dual-fuel (liquid or gas) nozzle is placed at the center of the swirler cup. The fuel from the nozzle impinges onto the venturi wall of the inner swirler. At the tip of the venturi, fuel was mixed with the counter-rotating swirling air from the inner and outer swirlers. The air-fuel mixture then was injected into the burner. Upstream of the swirler cup, a long diffuser was used to recover the pressure head and to reduce the air speed. A set of dilution holes were located downstream of the burner. The dilution jets were used to enhance the air-fuel mixing and reduce the equivalence ratio. Further downstream, another set of dilution holes were used to further reduce the gas temperature and to trim the outlet temperature profile. Besides the dilution holes, many cooling air holes were distributed in the burner to reduce the burner liner wall temperature. For a typical combustor design, about one third of the total air flow rate enters through the dome (swirl cup and dome cooling), and about one third of the total air goes through the dilution holes and the rest of air is used for combustor liner cooling. At one side of the burner, a quartz window was mounted for flow visualization. At the other side of the burner, a stainless steel plate, having the same size as the quartz window, was installed with four PCB dynamic pressure transducers, as shown in Fig. 3. The first transducer was used to measure the dynamic pressure inside the dome area. The second pressure transducer was used to measure the dynamic pressure near the first dilution air jet. The third transducer was located between the two sets of dilution holes and the fourth transducer was located around the second set of dilution holes.

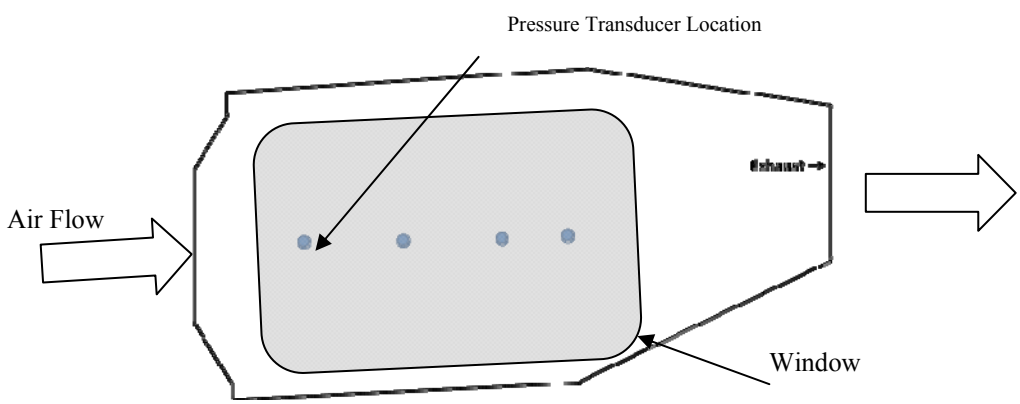


Figure 3: PCB dynamic pressure transducer locations

In addition to the flow meters, pressure transducers, and thermocouples that were used to monitor the tests, an AKG D112 dynamic cardioids microphone was positioned 304.8 mm (1 foot) away from the side of the burner to record the combustion acoustic emissions. A regular speed video camera and a PHANTOM V7.3 high-speed CCD camera were used to record the burning modes and flame dynamics. A type B thermocouple was used to monitor the center gas temperature at the combustor exit.

The experimental approach consisted of investigating the burning modes, acoustics emissions and combustor pressure dynamics as a function of equivalence ratio Φ, swirler pressure drop ratio $\Delta P/P$, and preheated inlet air temperature T_{inlet}. A typical measurement was initiated by setting $\Delta P = 2\%$ of atmosphere pressure, and setting T_{inlet} and Φ to the previously measured fuel lean blowout limit. The equivalence ratio was then continuously increased up to the stoichiometric equivalence ratio or fuel rich blow out by steps of $\Phi = 0.05$. The inlet air preheat temperature was then increased by steps of $\Delta T = 200$F and the procedure was repeated for the entire range of air–fuel ratios within the lean and rich limits. The tests were repeated at $\Delta P = 4\%$ and 6% of atmosphere pressure as well. All data were acquired with 10s test duration for each operating condition. At the same time, a regular video camera continuously monitored and recorded the combustion modes. The high speed CCD camera was used to record flame dynamics at some selected test conditions. The frame rate of the high speed camera was 6000 frames per second, which is sufficient to cover the frequency range of the combustion dynamics.

III. Results and discussion

The manifestation of combustion dynamics of the investigated combustor sector differs in many aspects .The flame structure, frequency spectra, and the relationship between combustion dynamics modes and flame structures are primarily dependent on the air-fuel ratio. In this section, we will first present data on flame and acoustic spectrum at different test conditions, and then discuss the relationship between the flame structures and the combustion dynamics modes.

A. Combustion dynamics modes, flow field and flame structures.

Figure 4 shows the acoustic energy spectra versus equivalence ratio, recorded by the AKG microphone, for an inlet air temperature of 600 F at a swirler pressure drop of 4% of atmosphere pressure. There are three frequency bands ($f_1 \approx 280$Hz, $f_2 \approx 420$Hz and $f_3 \approx 600$Hz) associated at different equivalence ratios It is concluded that the third acoustic mode is coincident with the quarter wave mode of the combustor chamber, which was calculated based on the combustion chamber length and the estimated bulk combustion gas temperature. This mode has weak acoustic energy while the other two modes have significant higher amplitudes. At low equivalence ratios, the combustion acoustic emission is the highest when Φ is around 0.32. The spectrum of the acoustic pressure recorded by the AKG microphone is shown in Figure 5, with the dominant frequency *f* at 280Hz. This low frequency mode (around 280Hz) occurs at the equivalence ratios from 0.25 to 0.38. Figure 6 shows the spectra from the four pressure transducers at different locations. The first pressure transducer, which is located in combustor dome zone, shows a result similar to that of the AKG microphone. A sharp peak is located at 280 Hz. With the second peak at 600 Hz. Higher frequency pressure oscillation can also be found in the first pressure transducer (Fig. 6). All four transducers contain the pressure oscillations at 280Hz. The higher frequency pressure oscillations decay quickly in the axial direction as shown on the spectra in the other three pressure transducers. It is clear that the high frequency modes are most likely related with the turbulence or local vortex structures in the dome zone and they do not form major acoustic emissions. When equivalence ratio increases, the acoustic energy peak moves to a slightly lower frequency. This may due to the lower air/fuel mixture temperature as more room temperature fuel is added to the mixture.

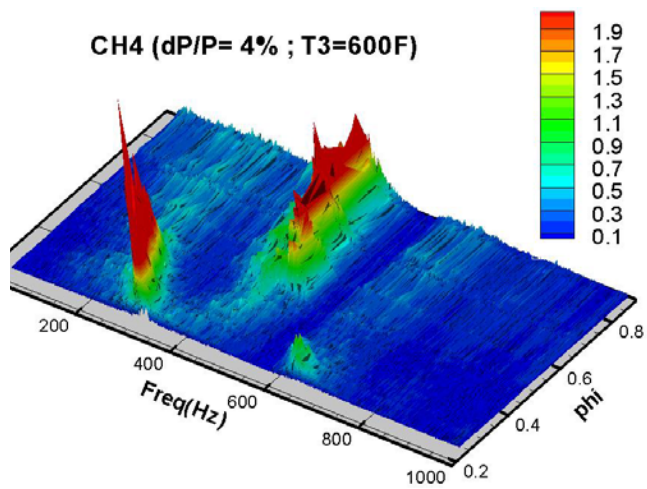


Figure 4, Dynamic energy iso-contour in parameter space(Φ, *f*)

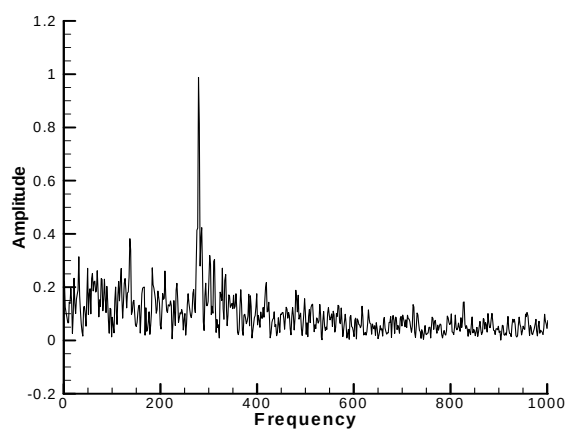


Figure 5, Frequency spectrum at low equivalence ratio recorded by AKG microphone

After the equivalence ratio rises above 0.38, the combustion abruptly transfers to a quiet combustion mode. The second acoustic mode was observed for an equivalence ratio of over 0.42 and higher, and it exhibited a 420Hz frequency (or higher, depending on the equivalence ratio). Figure 7 shows a typical frequency spectrum at the equivalence ratio $\Phi=0.62$ recorded by the AKG microphone. At this equivalence ratio, a dynamic mode, with a frequency of around 420 Hz, dominates the acoustic energy. The detailed pressure oscillations at different locations are presented in Figure 8. The dynamics pressure distribution inside the combustor is more complicated than the acoustics recorded by the microphone which is located 1 foot away from the combustor. Besides the 420 Hz peak recorded by the microphone, additional frequency peaks are observed near 420 Hz and at some higher frequencies.

Those dynamic modes, however, decay quickly when they propagate downstream. Like the previous case, these modes are related to the local turbulence and vortex flows, which do not form major acoustic emissions.

The knowledge of the flow field inside the combustor should be helpful in understanding the flame structure and its related acoustic mode. The combustor studied in this paper has a complicated flow field distribution that can cause the significant flame structure changes during various operating conditions. Figure 9 shows a mean velocity distribution inside the combustor, measured by an LDV (Bassam et al [12]). Near the swirler exit, a strong central recirculation zone was located in dome zone with high speed swirling flow attached to the dome wall. The swirling flow and the recirculation zone in the dome region play an important role in anchoring the flame. The unstable flow structures such as the vortex shedding and precessing vortex core (PVC) inside the swirling flow, however, could also cause combustion instability. Downstream of the dome zone, the primary dilution air jets are injected in to the combustor with high velocity. A low velocity wake region exists slightly downstream of the primary jet, and could be a possible location for flame anchoring. The unstable jet flow and the possible vortex shedding in the wake region may also induce unstable combustion. Because the flow field shows significantly different images at different regions inside the combustor, the flame structure will be entirely different, depending on where the flame is anchored.

To investigate the connections existing between acoustic modes and flame structures, the high speed camera pictures are essential to exhibit the combustion dynamics associated with the acoustic modes. The pictures obtained from high-speed video can be used to represent the locations of the flame structures corresponding to the different acoustic modes. Figures 10a and b show two instantaneous flame images associated with the first and second acoustic modes. For lean combustion conditions (first acoustic mode dominates), shown in Fig. 10 (a), combustion occurred mainly within the dome region. The flame was anchored at the swirler exit and most of the combustion was completed before the location of first set of dilution holes. No interaction happened between flame and dilution air. This combustion instability mode is strongly coupled with the swirling flow and central recirculation zone in the dome region. For the case with higher equivalence ratio (Fig. 10 b), the reactant mixture was so fuel rich that almost no combustion occurred in the dome region. The flame was anchored downstream of the dilution holes. At this location, the flame front interacts with the dilution air jet and the vortex structure in the wake region. The second acoustic mode could be attributed to such strong interaction between the flame front and the aerodynamics.

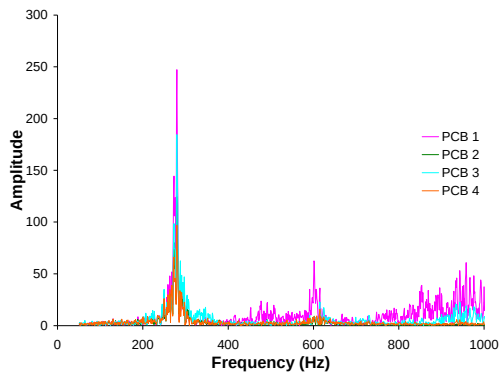


Figure 6, Frequency spectra at low equivalence ratio recorded by different locations' pressure transducers

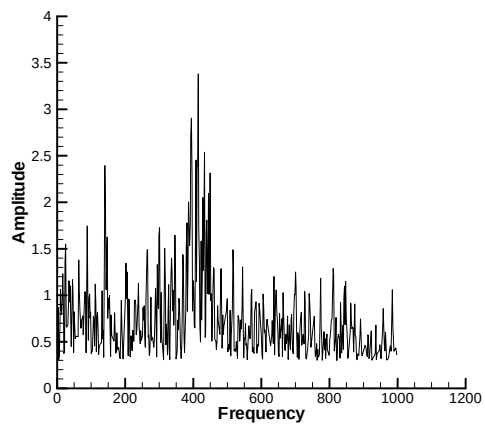


Figure 7, Frequency spectrum at high equivalence ratio recorded by AKG microphone

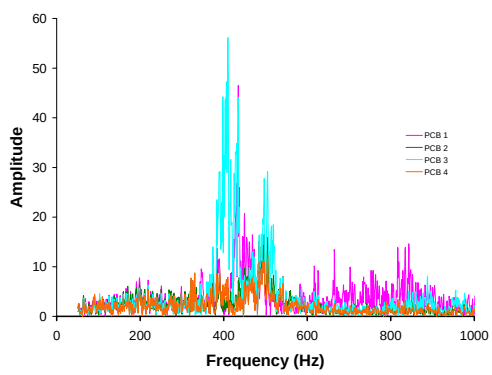


Figure 8, Frequency spectra at low equivalence ratio recorded by different locations' pressure transducers

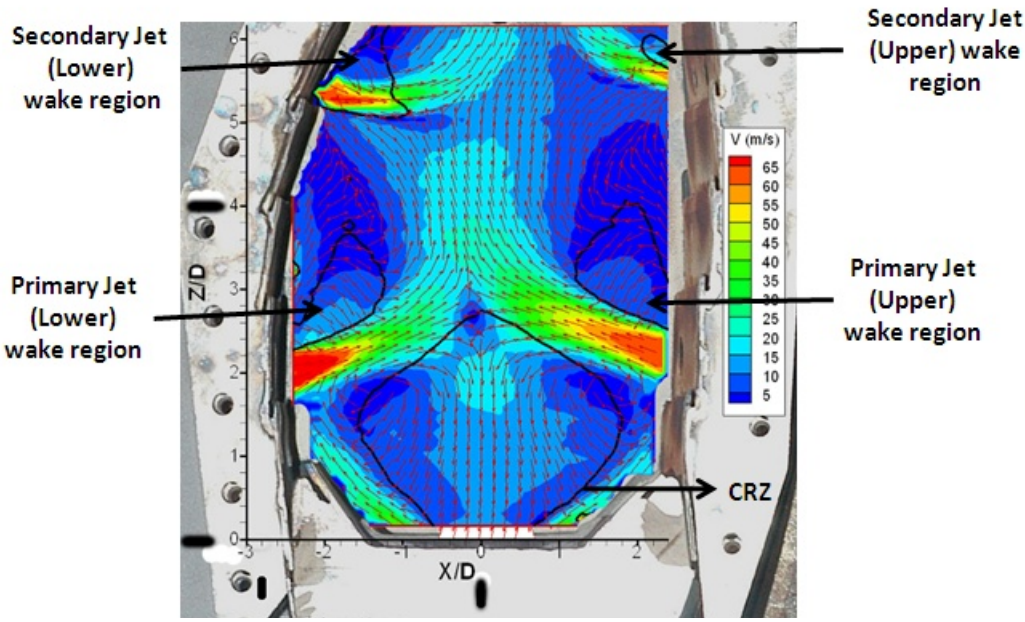


Figure 9, Combustor total velocity contours and velocity vectors measured by LDV

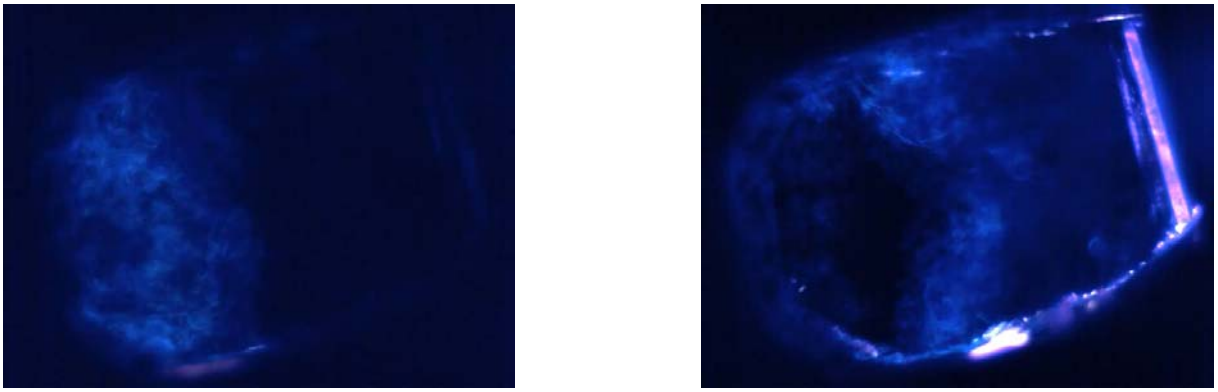


Figure 10.(a) Flame structure associated with first acoustic mode. (b) Flame structure associated with the second acoustic mode

B. Flame dynamics based on the FFT analysis on high speed videos

To further investigate the acoustic characteristics of the burner, a video FFT software was developed to compute the local energy release rate (amplitude and phase) distribution based on the chemiluminescence recorded by the high speed camera (Fumi et al[13,14]). Figure 11 shows the spatially distributed energy release rate associated with the first acoustic mode (f=280Hz). Figure 11(a) shows that high energy release rate is concentrated at two locations near the swirler exit and then gradually dissipates downstream. The frequency of the high energy release flame oscillations matches the high amplitude acoustic emission frequency. This indicates the acoustic energy source comes from the flame front oscillation which is located at the strong shear layer between outer swirling jet and recirculation zone. Figure 11b shows the phase angle distribution inside the burner. The phase angle difference between top and bottom high acoustic energy spots is about 180 degrees (3.14 radian). The 180 phase angle difference of these two opposite locations indicates a circumferential energy release rate related to this

combustion dynamics. Since the swirling flow dominates in the combustor dome region, the interaction between the flame and the unstable swirling flow seems to be the mechanism that generates the first acoustic mode. In the region between the two high acoustic energy spots, the swirling flow produces opposite motion at the front and back of the combustor, resulting in low phase angle differences.

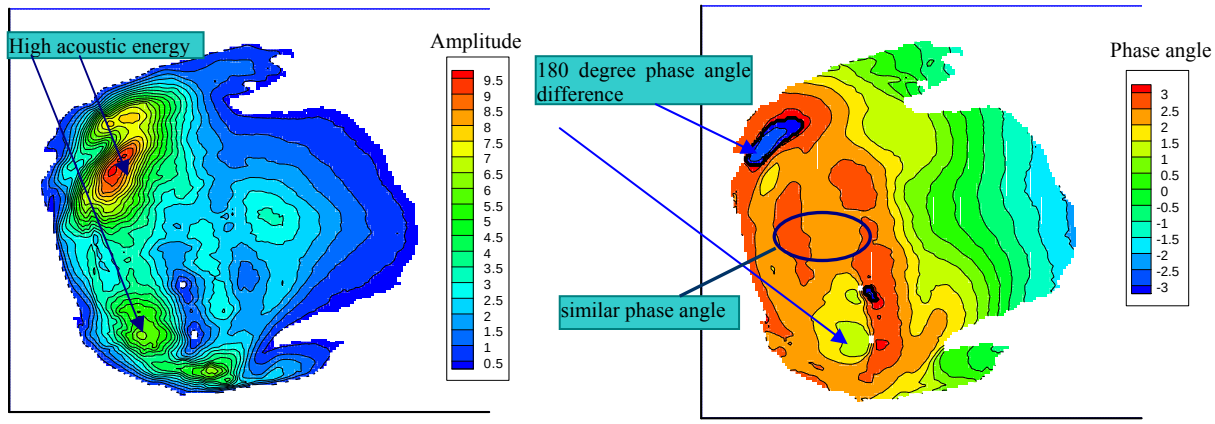


Figure 11 (a) First acoustic mode energy distribution. (b) First acoustic mode phase angle distribution

Figure 12 shows the acoustic energy and related phase angle distribution associated with the second acoustic mode (f=414Hz) for combustion at Φ=0.61. Unlike previous combustion case, figure 12(a) shows that strong acoustic energy spots are concentrated around the dilution air jets. This location also coincides with the flame front location showed in Fig. 10(b). Figure 12(b) shows the phase angle distribution inside the burner. No circumferential acoustic mode was found in this case. On the other hand, there is an acoustic wave propagating around the dilution air jet. The figure shows there is a 360 degree phase angle difference between the exit of the dilution hole and the location where two dilution air jets meet. This phase angle distribution reveals that the unsteady combustion mechanism of this acoustic mode is totally different to the previous case. It is more likely that this acoustic mode was driven by the interaction between the flame and dilution air jets.

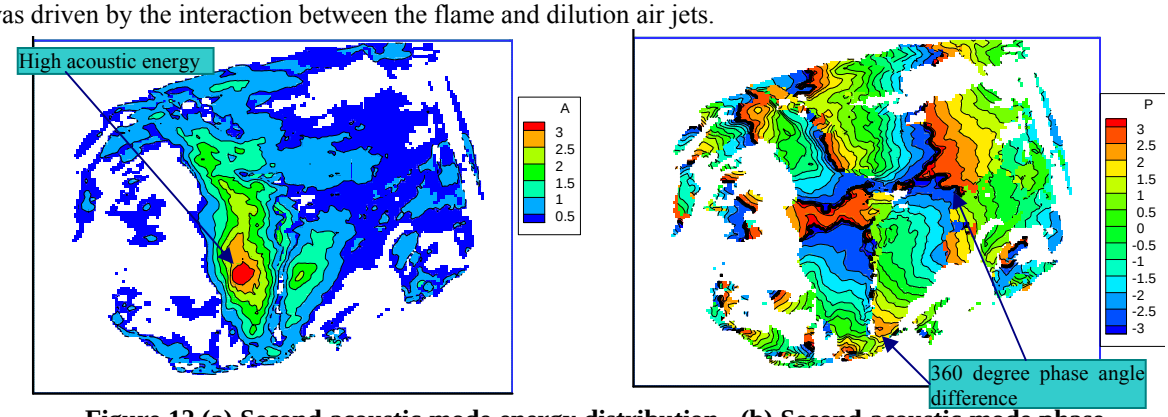


Figure 12.(a) Second acoustic mode energy distribution . (b) Second acoustic mode phase angle distribution

C. The effect of the inlet air temperature to the combustion dynamics

Figure 13 shows the effect of inlet air temperature to the combustion dynamics. The tests were performed at a swirler pressure drop of 4% of atmospheric pressure. Figure 13 presents the acoustic modes observed for air inlet temperatures of 200 F, 400 F and 600 F. Both first and second acoustic modes were observed for all three inlet air temperature cases. With increase of the inlet air temperature, both acoustic modes were observed to shift to higher frequencies. This kind of shift is predictable since the increase of the air temperature raises the flame temperature which results in an increase in the premixed burning speed, convective speed and acoustic velocity.

D. The effect of the pressure drop to the combustion dynamics

Figure 14 shows the effect of the pressure drop across the swirler cup to the combustion dynamics. Three pressure drops of 2%, 4% and 6% of atmospheric pressure were chosen. The inlet temperature was 600F for all three pressure drops. Both acoustic modes were observed for all three test conditions. But the effects of pressure drops on the two acoustic modes are different. Figure 14(a) shows the effect of pressure drop on the first acoustic mode (Φ=0.32 for all three pressure drops). With an increase in the pressure drop, the first mode acoustics moves to higher frequency. The acoustic frequency is observed to be proportional to the convective air speed. This again indicates the first acoustic mode is tightly connected to the characteristics of the swirling flow. Unlike the first acoustic mode, figure 14(b) shows that the effects of the pressure drop on the second acoustic mode (Φ=0.66 for all three pressure drops) are minor for 4% and 6% pressure drop. The physical processes that govern this behavior are still under investigation.

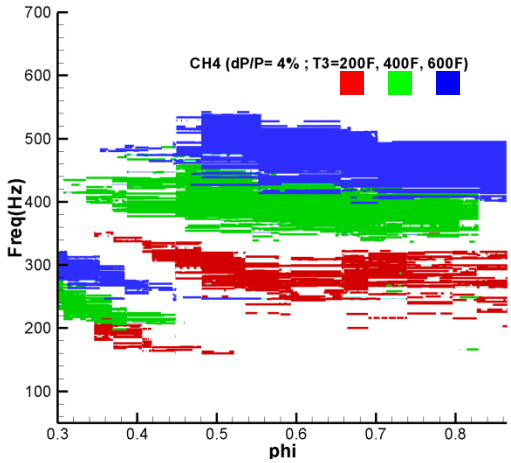


Figure 13. Acoustics frequencies modes at different inlet air temperatures

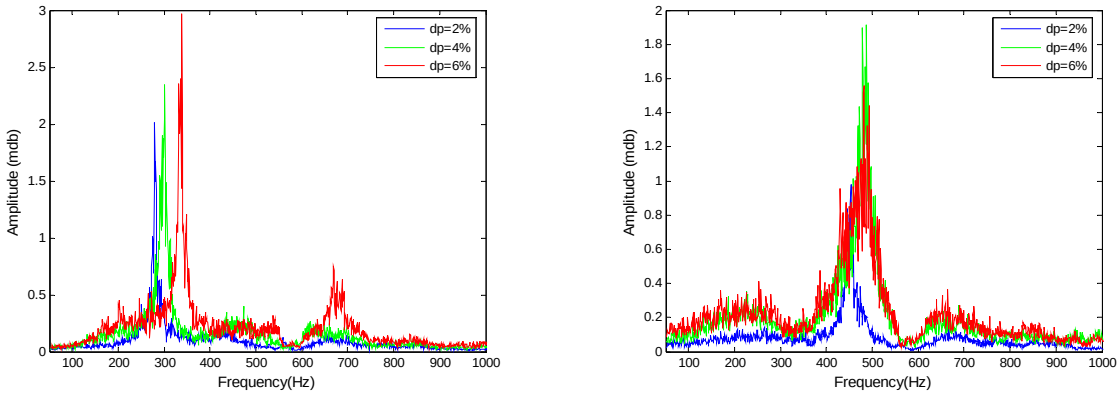


Figure 14 (a) Frequency spectrum of first acoustic mode under different pressure drops. (b) Frequency spectrum of second acoustic mode under different pressure drops.

IV. Conclusion

The acoustic emissions and flame dynamics of an atmospheric swirl-stabilized burner fed with natural gas has been investigated experimentally. This study revealed the existence combustion dynamics from both fuel-lean and fuel rich domes. The fuel-lean dome dynamics was associated with a flame structure which is anchored at the swirler exit and the energy release was completed within the dome. As equivalence ratio was increased, the flame front will propagate downstream. The fuel-rich dome combustion dynamics occurred when the primary flame front was anchored downstream of the primary dilution jets. The inlet air temperatures and air pressure drops do not change the combustion dynamics modes but they do affect the acoustic emission frequencies. The physical processes govern the acoustic frequency changes are still under investigation.

High speed video has been shown to be a powerful tool to analyze combustion instability. The images from the high speed video reveal the flame structures associated with a particular acoustic mode. Applying FFT analysis on the images reveals interactions among the energy release rate, the flame motion and the acoustic modes. The FFT analysis of the fuel-lean mode dynamics shows that a circular oscillation mechanism related to this mode, which is strongly coupled with the unsteady swirling flow. While the fuel-rich mode dynamics is related to an interaction of the unstable dilution air jet with the unburned fuel-rich dome flows.

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